

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (B.Sc.I.T Semester V)**  
**Data Visualization with Power BI and Tableau**

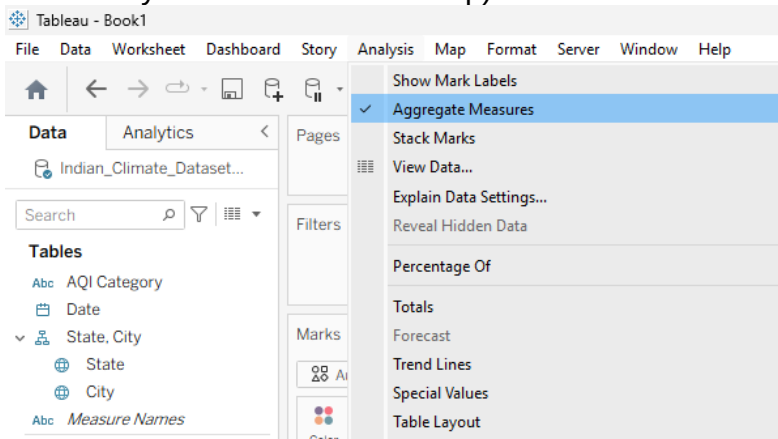
## Practical - VIII

|                             |                                  |
|-----------------------------|----------------------------------|
| Roll No.: T032              | Name: KHUSHI                     |
| Class: TYIT                 | Batch: 2                         |
| Date of Assignment: 27-8-26 | Date/Time of Submission: 27-8-26 |

### A: Add and Interpret Trend Lines (Linear, Exponential) :

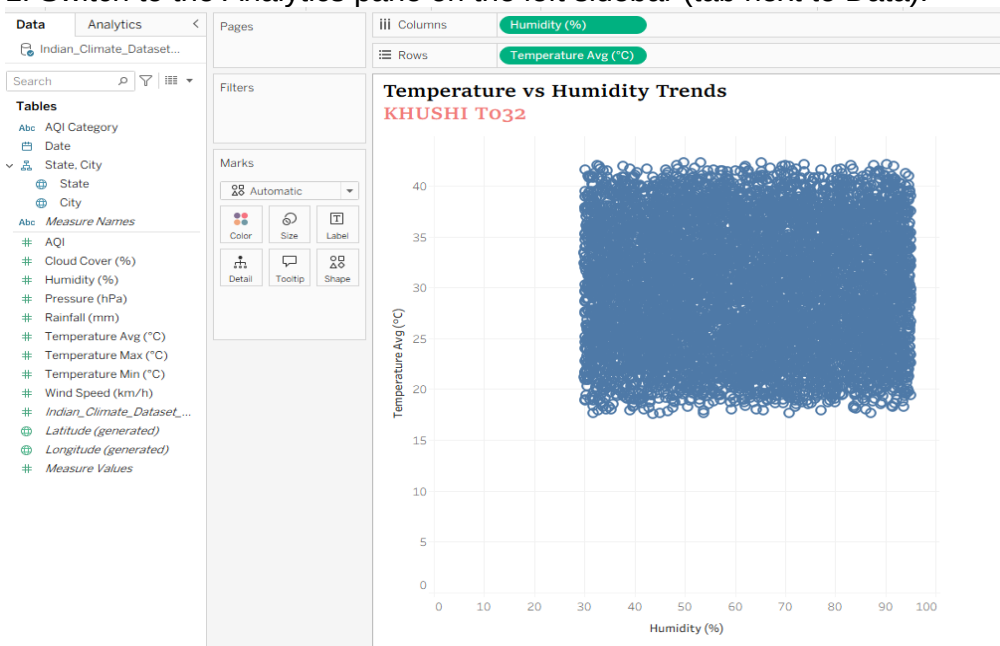
#### 1. Connect Dataset & Build a Scatter Plot / Line Chart

1. Open Tableau Desktop and load Indian\_Climate\_Dataset\_2024\_2025.csv.
2. Open a new worksheet and rename it to Temperature vs Humidity Trends.
3. Drag Humidity (%) to the Columns shelf.
4. Drag Temperature\_Avg (°C) to the Rows shelf.
5. In the top menu, go to Analysis and uncheck Aggregate Measures (so each data point is plotted individually rather than summed up).

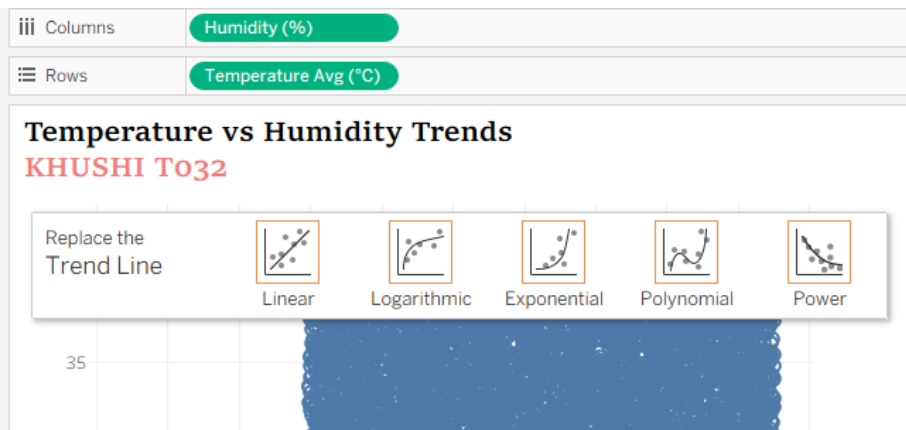


#### a Linear Trend Line

1. Switch to the Analytics pane on the left sidebar (tab next to Data).



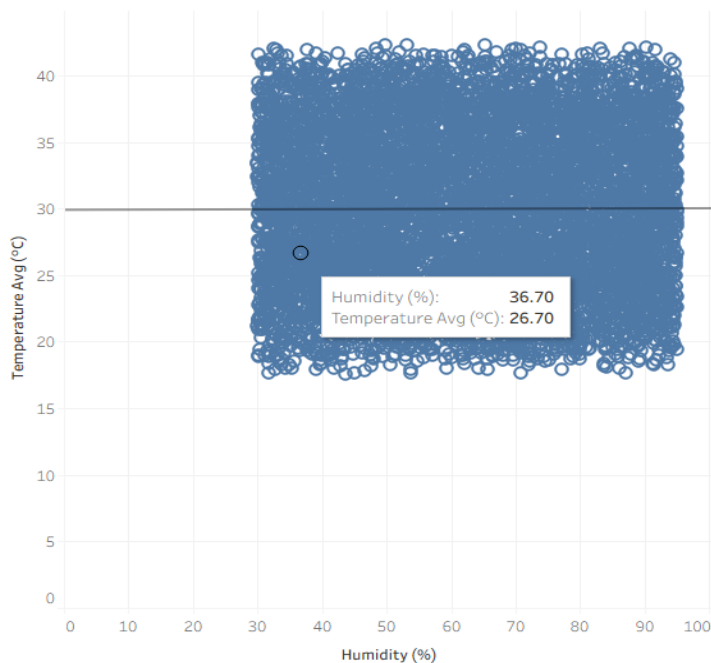
2. Under Model, drag Trend Line onto the canvas and drop it onto Linear.



3. Hover over the trend line to observe the statistical popup:

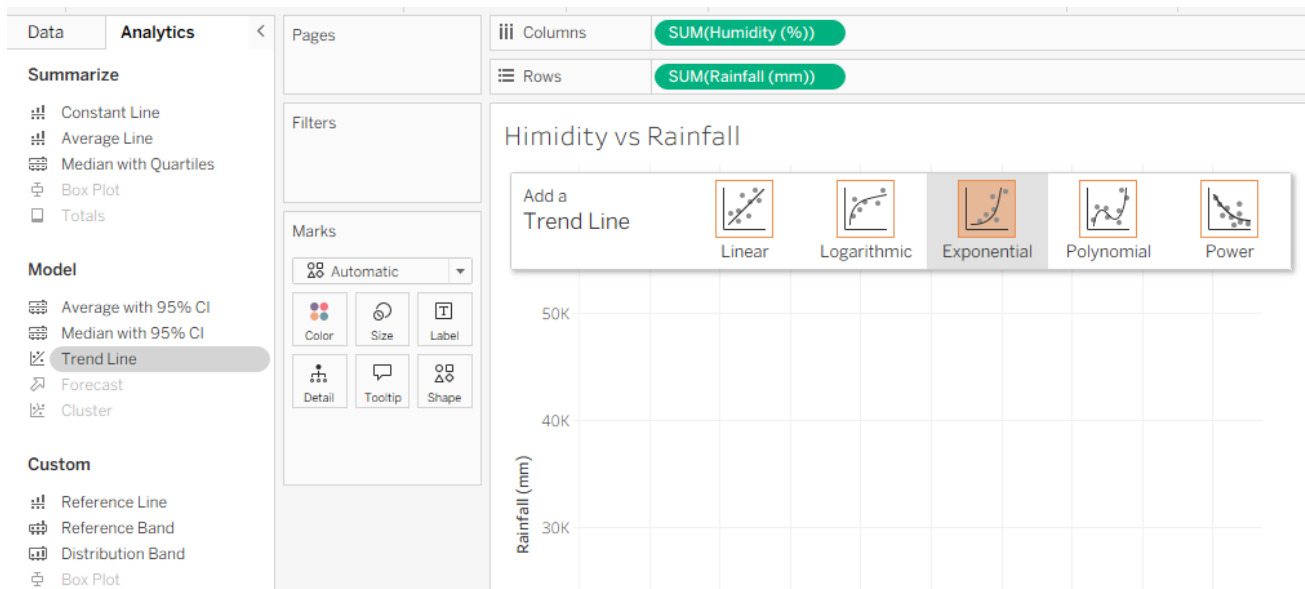
- It displays the linear equation:  $\text{Temperature\_Avg} = a \times \text{Humidity} + b$
- It displays  $R^2$  (coefficient of determination) and P-value (statistical significance).

**Temperature vs Humidity Trends**  
KHUSHI T032

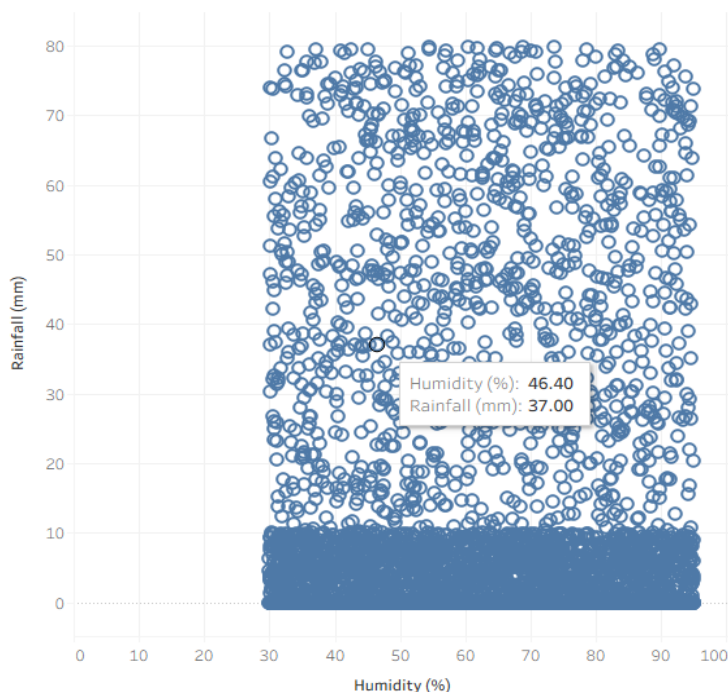


3. Add an Exponential Trend Line for Non-Linear Measures (Rainfall vs. Humidity)

1. Open a new worksheet named Humidity vs Rainfall.
2. Drag Humidity (%) to Columns and Rainfall (mm) to Rows.
3. Uncheck Analysis  $\diamond$  Aggregate Measures.
4. From the Analytics pane, drag Trend Line onto Exponential (or right-click the canvas  $\diamond$  Trend Lines  $\diamond$  Show Trend Lines  $\diamond$  Edit Trend Lines and change the model to Exponential).



**Humidity vs Rainfall**  
**KHUSHI T032**



#### 4. Interpret the Results

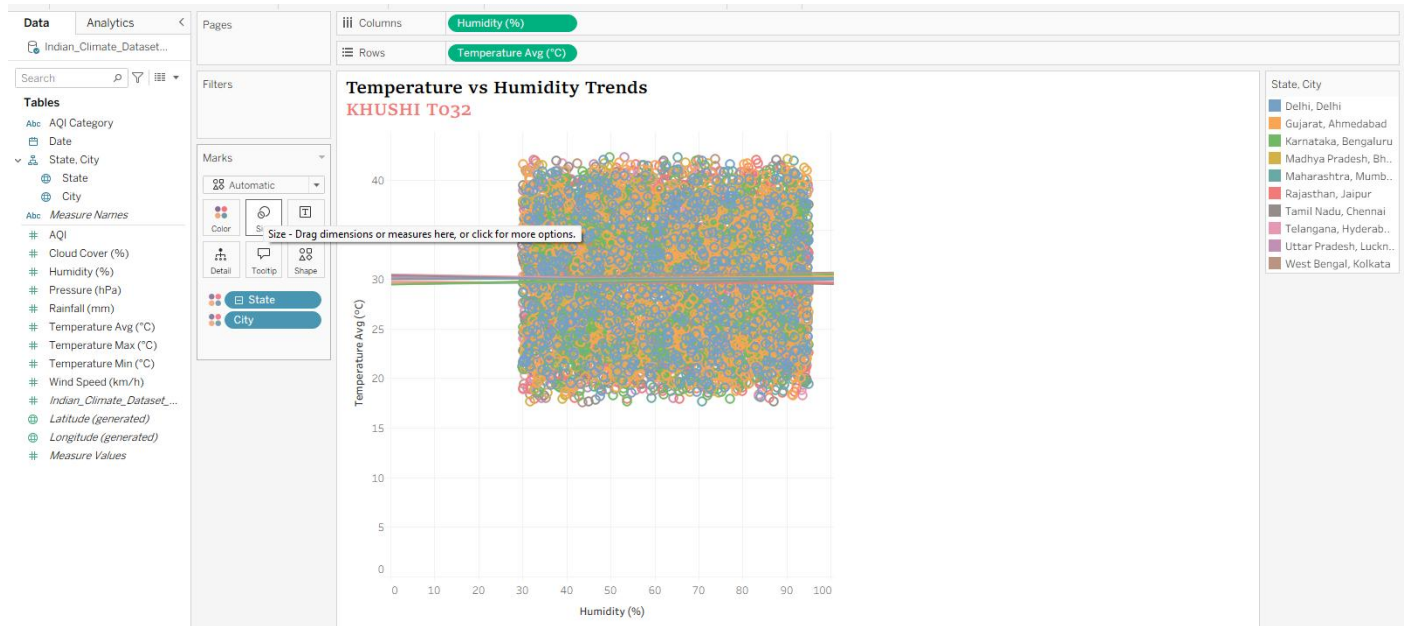
- Linear Model: Describes direct proportional/inverse relationships (e.g., checking if higher humidity steadily decreases or increases average temperature).
- Exponential Model: Captures accelerating relationships (e.g., how rainfall spikes exponentially once humidity crosses higher thresholds).

#### **B: Customize and Analyze Trend Models :**

##### 1. Disaggregate Trend Lines by City

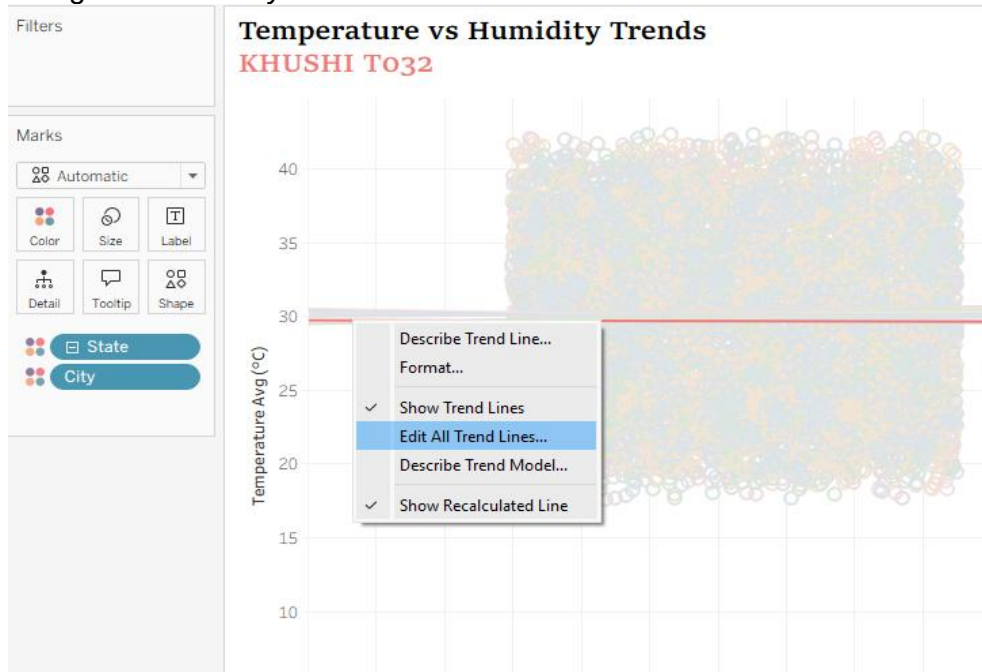
1. Open your Temperature vs Humidity Trends sheet from Part A (or create a duplicate).
2. Drag City from the Data pane onto Color on the Marks card.
  - o Notice what happens: Instead of one global trend line, Tableau automatically computes 10 distinct trend lines, one for each city!

3. Observe how coastal cities (like Mumbai or Chennai) show different slopes compared to inland cities (like Delhi or Jaipur).



## 2. Compare Different Model Types (Polynomial & Logarithmic)

1. Right-click on any trend line in the view and select Edit Trend Lines...



2. In the options dialog, experiment with switching the Model Type:

- o Linear: Assumes a constant rate of change.
- o Logarithmic: Useful when values rise rapidly at first and then level off.
- o Polynomial (Degree 2 or 3): Great for U-shaped or S-shaped seasonal curves (e.g., temperatures dipping mid-year and rising again).

3. Select Polynomial (Degree 2) and click OK to observe how well a curved line fits seasonal fluctuations.

## Trend Lines Options

## Model Type

- ☒ Linear  
☐ Logarithmic  
☐ Exponential  
☐ Power  
☐ Polynomial Degree: 3

## Factors

Build separate trend lines based on the following dimensions:

- ☒ City  
☒ State

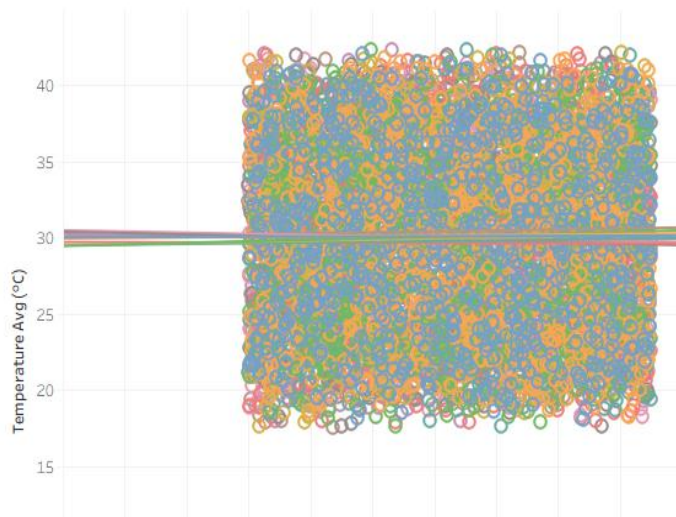
## Options

- ☒ Show tooltips  
☐ Show confidence bands  
☒ Allow a trend line per color  
☒ Show recalculated line for highlighted or selected data points  
☐ Force y-intercept to zero

OK

## Temperature vs Humidity Trends

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## Trend Lines Options

## Model Type

- ☐ Linear  
☒ Logarithmic  
☐ Exponential  
☐ Power  
☐ Polynomial Degree: 3

## Factors

Build separate trend lines based on the following dimensions:

- ☒ City  
☒ State

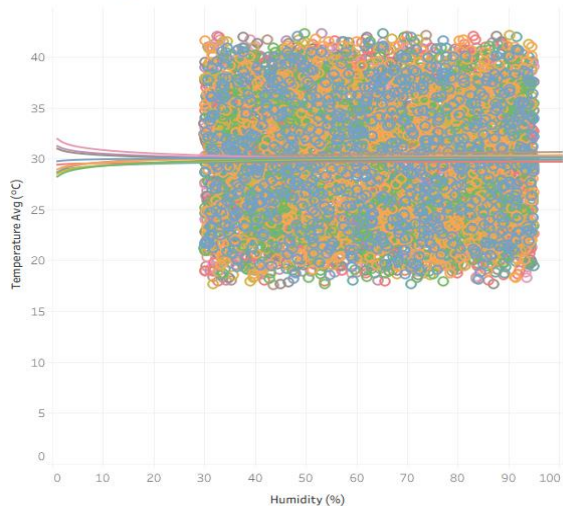
## Options

- ☒ Show tooltips  
☐ Show confidence bands  
☒ Allow a trend line per color  
☒ Show recalculated line for highlighted or selected data points  
☐ Force y-intercept to zero

OK

## Temperature vs Humidity Trends

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## Trend Lines Options

## Model Type

- ☐ Linear  
☐ Logarithmic  
☒ Exponential  
☐ Power  
☐ Polynomial Degree: 3

## Factors

Build separate trend lines based on the following dimensions:

- ☒ City  
☒ State

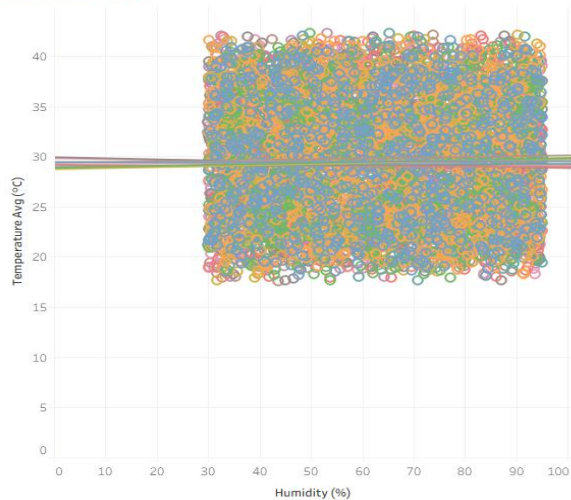
## Options

- ☒ Show tooltips  
☐ Show confidence bands  
☒ Allow a trend line per color  
☒ Show recalculated line for highlighted or selected data points  
☐ Force y-intercept to zero

OK

## Temperature vs Humidity Trends

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## Trend Lines Options

## Model Type

- ☐ Linear  
☐ Logarithmic  
☐ Exponential  
☐ Power  
☒ Polynomial Degree:

## Factors

Build separate trend lines based on the following dimensions:

- ☒ City  
☒ State

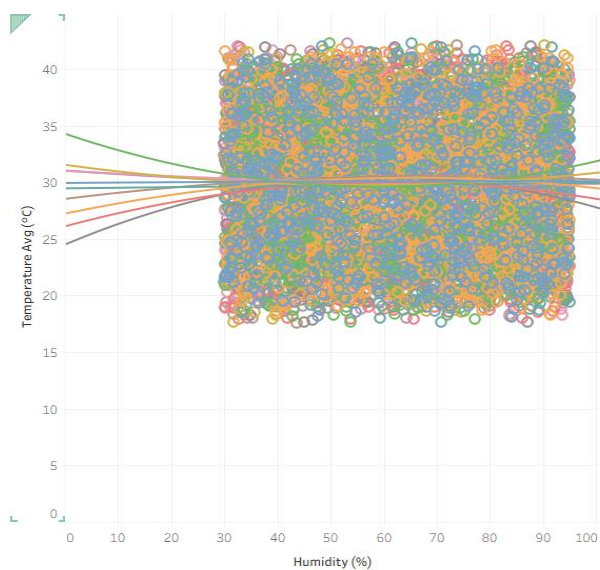
## Options

- ☒ Show tooltips  
☐ Show confidence bands  
☒ Allow a trend line per color  
☒ Show recalculated line for highlighted or selected data points  
☐ Force y-intercept to zero

OK

## Temperature vs Humidity Trends

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## 3. Access Full Statistical Summaries (Describe Model)

- Right-click anywhere on the chart background or trend line and select Describe Trend Line...
  - This shows a quick overview of formula equations, R<sup>2</sup> and P-value per city.

## Describe Trend Line

P-value: 0.366146

Equation: Temperature Avg (°C) = -0.00101407\*Humidity (%)^2 + 0.125697\*Humidity (%) + 26.1069

## Coefficients

| Term           | Value      | StdErr    | t-value  | p-value  |
|----------------|------------|-----------|----------|----------|
| Humidity (%)^2 | -0.0010141 | 0.0007159 | -1.41646 | 0.157069 |
| Humidity (%)   | 0.125697   | 0.0901763 | 1.3939   | 0.163773 |
| Intercept      | 26.1069    | 2.65435   | 9.83551  | < 0.0001 |

Copy

Describe Trend Line...

Format...

✓ Show Trend Lines

Edit All Trend Lines...

Describe Trend Model...

✓ Show Recalculated Line

- Next, right-click and select Describe Trend Model...

o A detailed window will pop up showing the ANOVA table, Degrees of Freedom, Sum of Squared Errors (SSE), and F-statistic.

Describe Trend Line...

Format...

✓ Show Trend Lines

Edit All Trend Lines...

Describe Trend Model...

✓ Show Recalculated Line

## Describe Trend Model

Trend Lines Model

A polynomial trend model of degree 2 is computed for Temperature Avg (°C) given Humidity (%).

Model formula:  $\text{City} * \text{State} * (\text{Humidity} (\%)^2 + \text{Humidity} (\%) + \text{intercept})$   
 Number of modeled observations: 7310  
 Number of filtered observations: 0  
 Model degrees of freedom: 30  
 Residual degrees of freedom (DF): 7280  
 SSE (sum squared error): 259327  
 MSE (mean squared error): 35.6218  
 R-Squared: 0.0033993  
 Standard error: 5.9684  
 p-value (significance): 0.686762

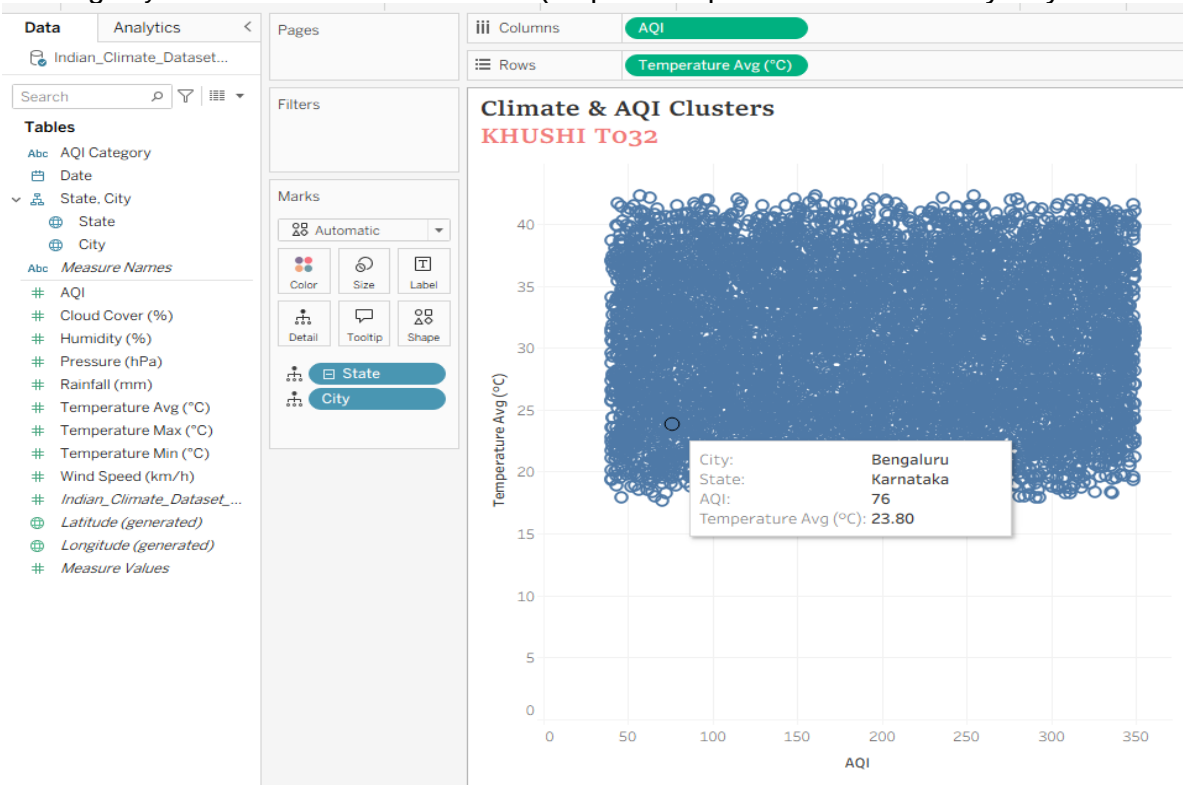
## Analysis of Variance:

| Field | DF | SSE | MSE | F | p-value |
|-------|----|-----|-----|---|---------|
|       |    |     |     |   |         |

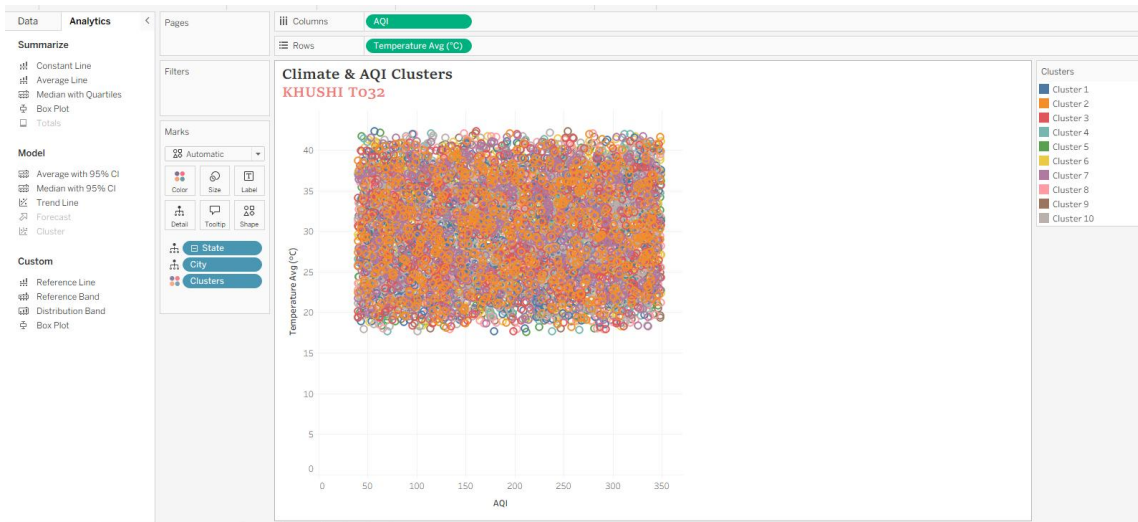
Copy

**C: Perform Clustering on Datasets :****1. Build a Multi-Metric Scatter Plot Matrix**

1. Open a new worksheet and name it Climate & AQI Clusters.
2. Drag AQI to Columns.
3. Drag Temperature\_Avg (°C) to Rows.
4. In the top menu, go to Analysis and uncheck "Aggregate Measures".
5. Drag city to Detail on the Marks card (so points represent individual city-day observations).

**2. Apply Built-in Tableau Clustering**

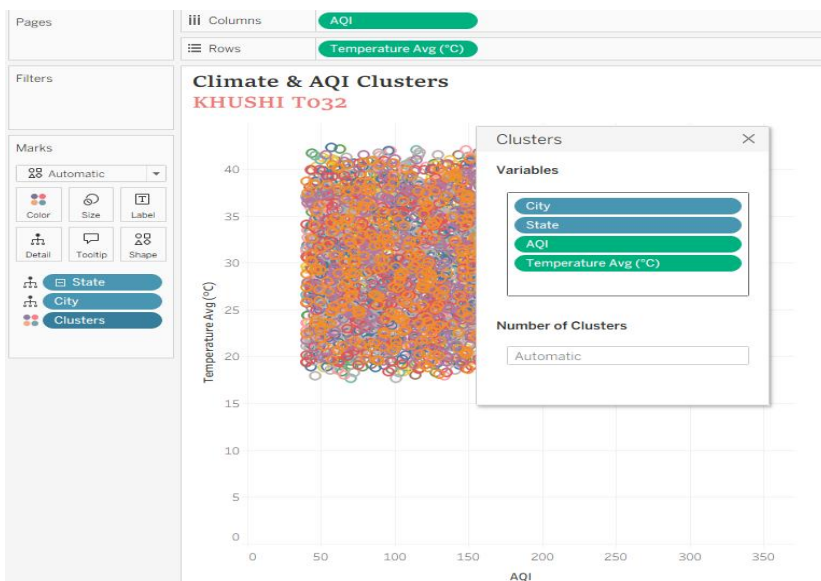
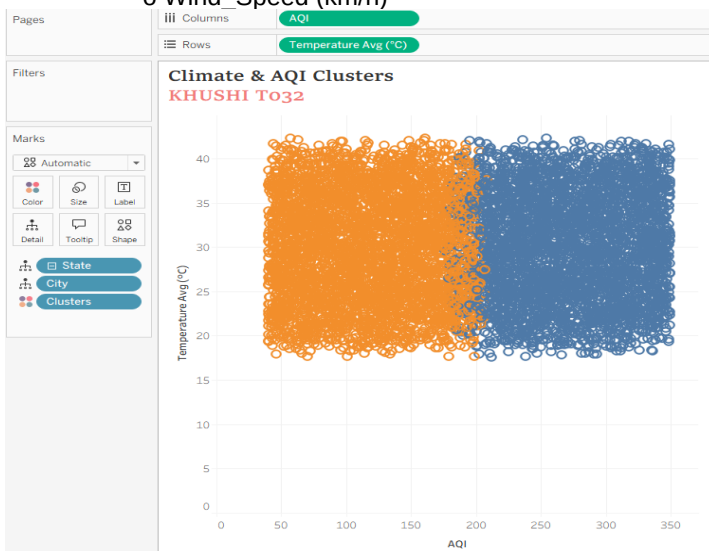
1. Switch to the Analytics pane on the left sidebar.
2. Under Model, drag Cluster directly onto the canvas.



3. A Clusters dialog box will appear on the right side of the view.

4. Drag additional variables into the Variables box to create a multi-variable cluster model (If you can't find Variable box Right Click Clusters in Marks Card, click on Edit Clusters and drag and drop the variables on the box):

- o AQI
- o Temperature\_Avg (°C)
- o Humidity (%)
- o Wind\_Speed (km/h)

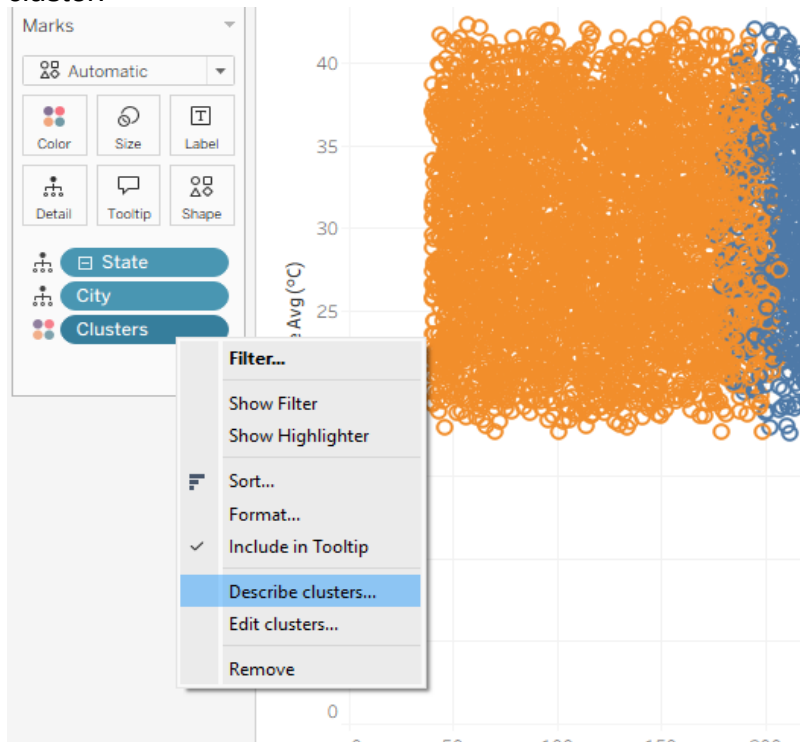


5. Under Number of Clusters, leave it as Automatic (Tableau will calculate the optimal number using K-means), or type 3 or 4 to explicitly create 3 or 4 distinct climate groups.
6. Close the cluster popup window. Tableau will color-code every data point by its assigned cluster (e.g., Cluster 1, Cluster 2, Cluster 3).

### 3. Analyze Cluster Profiles & Save as Dimension

1. Right-click on the Clusters field in the Marks card (or legend) and select Describe Clusters...
2. In the popup window, review two key tabs:

o **Summary:** Shows the mean value of each metric (AQI, Temperature, Humidity, Wind Speed) for each cluster.



Describe Clusters

Summary Models

**Inputs for Clustering**

Variables: City, State, Sum of AQI, Sum of Humidity (%), Sum of Temperature Avg (°C), Sum of Wind Speed (km/h)

Level of Detail: City, State

Scaling: Normalized

**Summary Diagnostics**

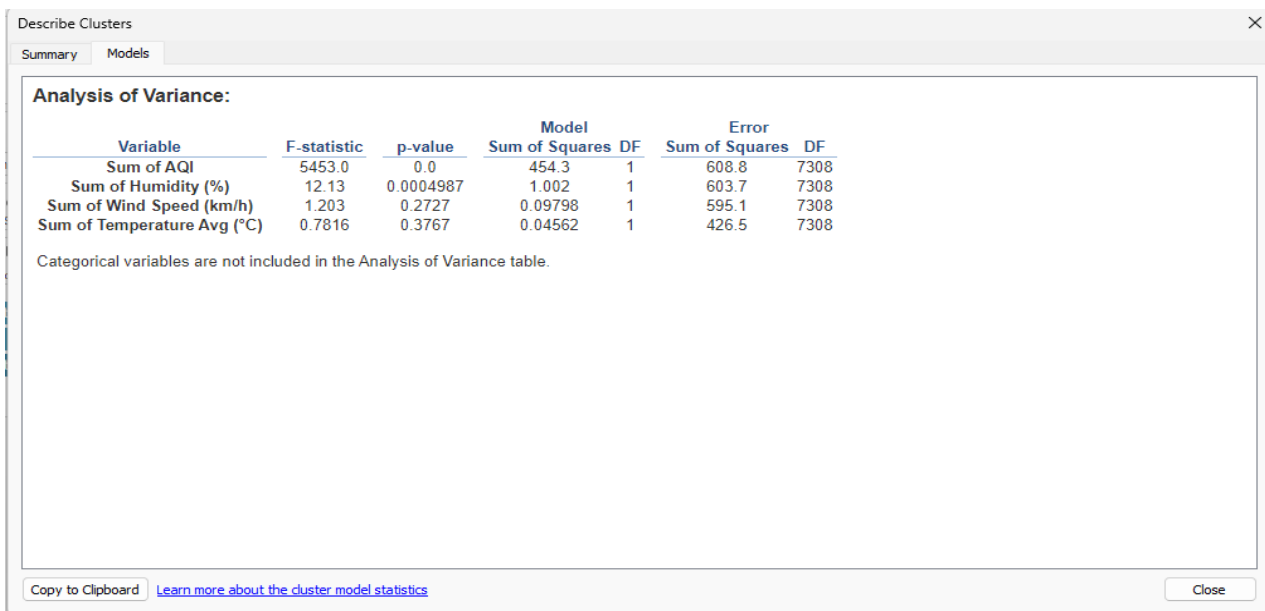
Number of Clusters: 2  
 Number of Points: 7310  
 Between-group Sum of Squares: 459.75  
 Within-group Sum of Squares: 6566.3  
 Total Sum of Squares: 7026.1

| Clusters  | Number of Items | Sum of AQI | Sum of Humidity (%) | Centers                     |                          | Most Common |             |
|-----------|-----------------|------------|---------------------|-----------------------------|--------------------------|-------------|-------------|
|           |                 |            |                     | Sum of Temperature Avg (°C) | Sum of Wind Speed (km/h) | City        | State       |
| Cluster 1 | 3636            | 271.19     | 61.889              | 30.042                      | 13.438                   | Jaipur      | Telangana   |
| Cluster 2 | 3674            | 117.13     | 63.411              | 29.919                      | 13.607                   | Hyderabad   | West Bengal |

☐ Show scaled centers

Copy to Clipboard [Learn more about the cluster summary statistics](#) Close

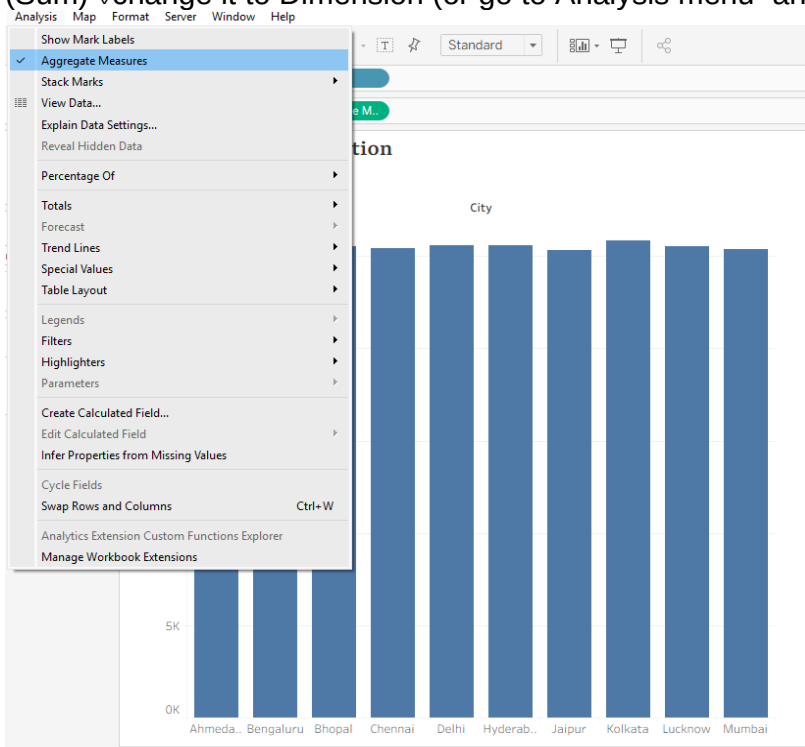
o **Models:** Shows ANOVA statistics (F-statistics and p-value) indicating which variables contributed most heavily to separating the clusters.



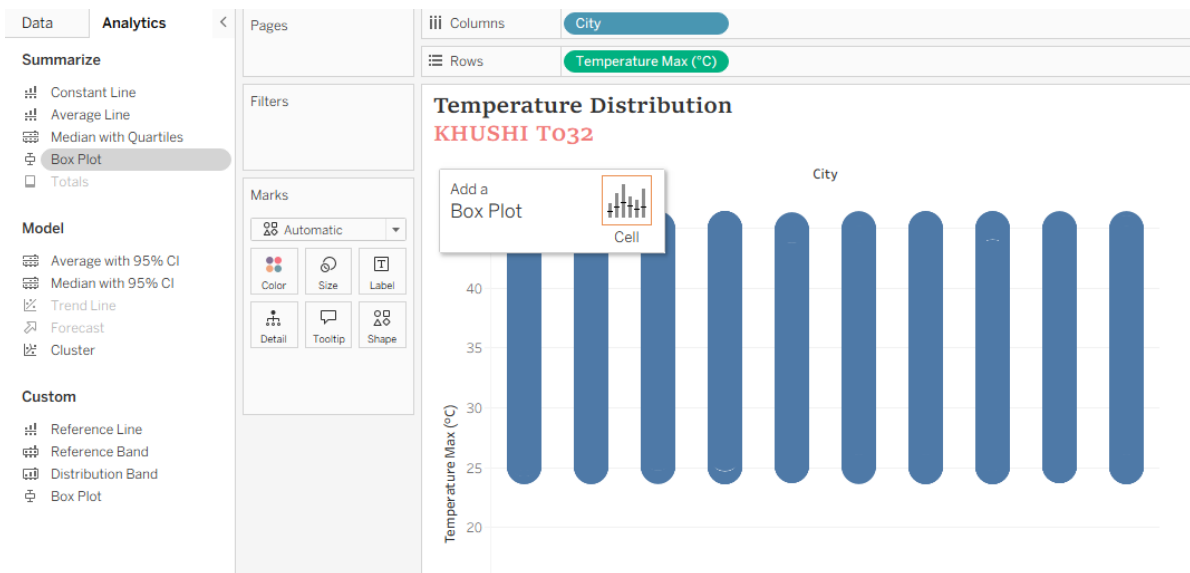
## D: Analyze Distributions using Built-in Tools :

### 1. Create a Box Plot for Temperature Distributions by City

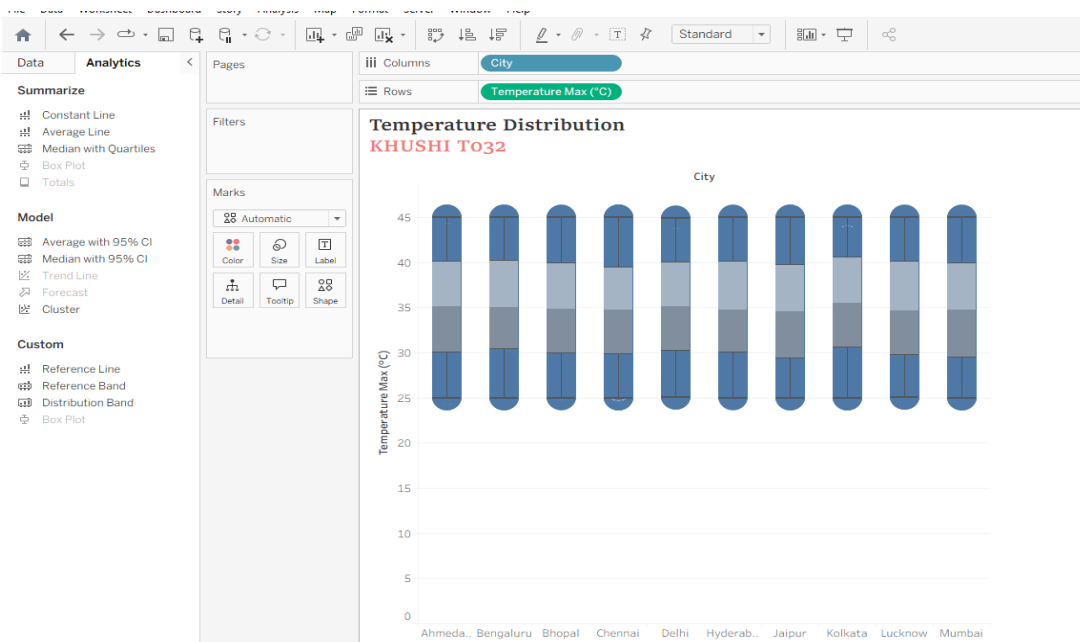
1. Open a new worksheet and name it Temperature Distributions.
2. Drag city to Columns.
3. Drag Temperature\_Max (°C) to Rows.
4. Tableau will default to SUM(Temperature\_Max). Right-click the Temperature\_Max pill on Rows  $\diamond$  Measure (Sum)  $\diamond$  change it to Dimension (or go to Analysis menu and uncheck "Aggregate Measures").



5. Switch to the Analytics pane on the left sidebar.
6. Under Custom, drag Box Plot onto the canvas and drop it on the chart.



7. Hover over any box plot to inspect key statistical metrics:
- o Upper Whisker / Maximum: Upper fence boundary.
  - o Upper Hinge (Q3): 75th percentile.
  - o Median (Q2): 50th percentile.
  - o Lower Hinge (Q1): 25th percentile.
  - o Outliers: Individual points plotted beyond the whiskers

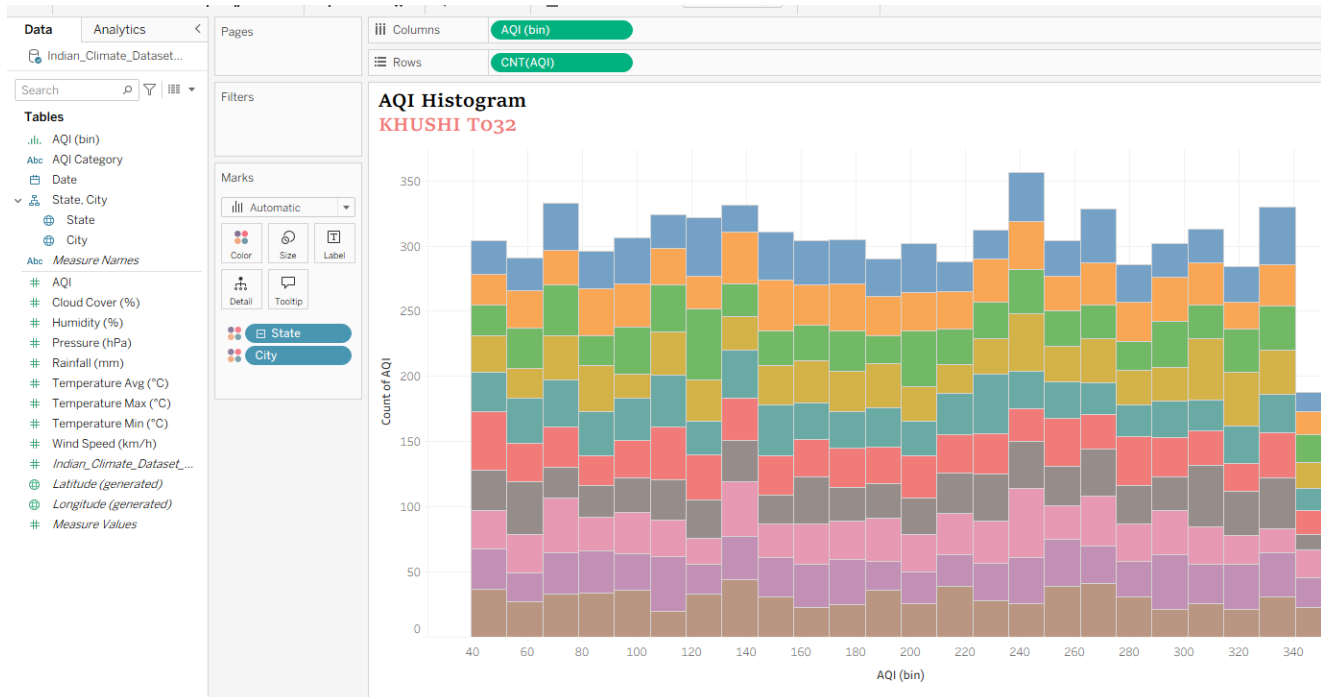


## 2. Build an AQI Histogram & Quantile/Percentile View

1. Open a new worksheet named AQI Histogram.
2. From the Data Pane, click on AQI.
3. Open the Show Me panel in the top-right corner of Tableau.
4. Select the Histogram icon.

(Tableau will automatically create an AQI (bin) dimension and place AQI (bin) on Columns and CNT(AQI) on Rows).

5. Drag City to Color on the Marks card to see how air quality distributions stack across cities.



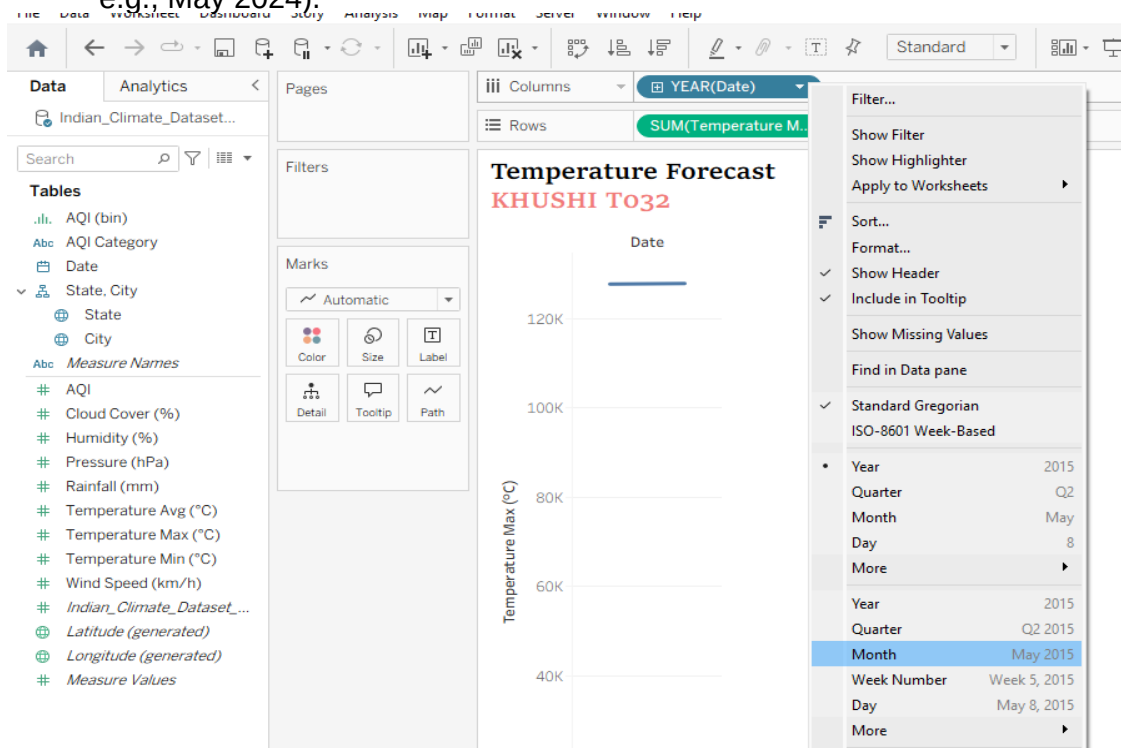
## E: Apply Forecasting Techniques :

### 1. Set Up a Time Series Line Chart

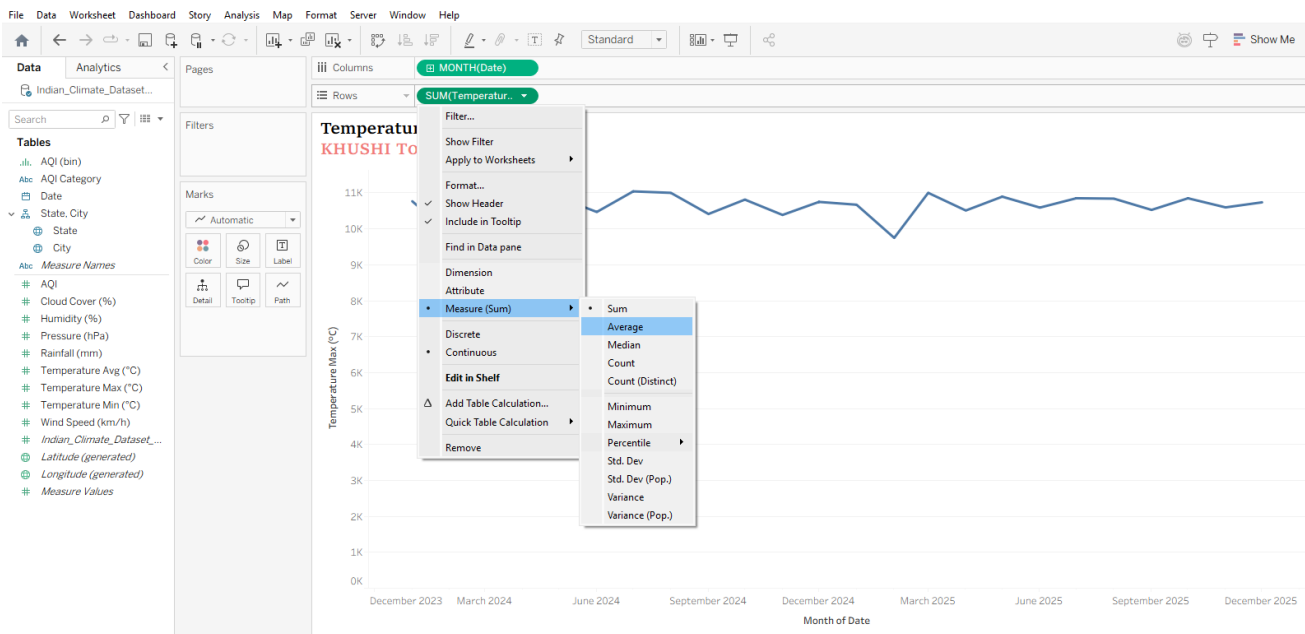
1. Open a new worksheet and name it Temperature Forecast.

2. Drag Date to Columns.

o Note: Tableau will default to YEAR(Date). Click the dropdown arrow on the Datepill and change it to continuous Month (the second Month option in the menu, which displays as Month Year, e.g., May 2024).



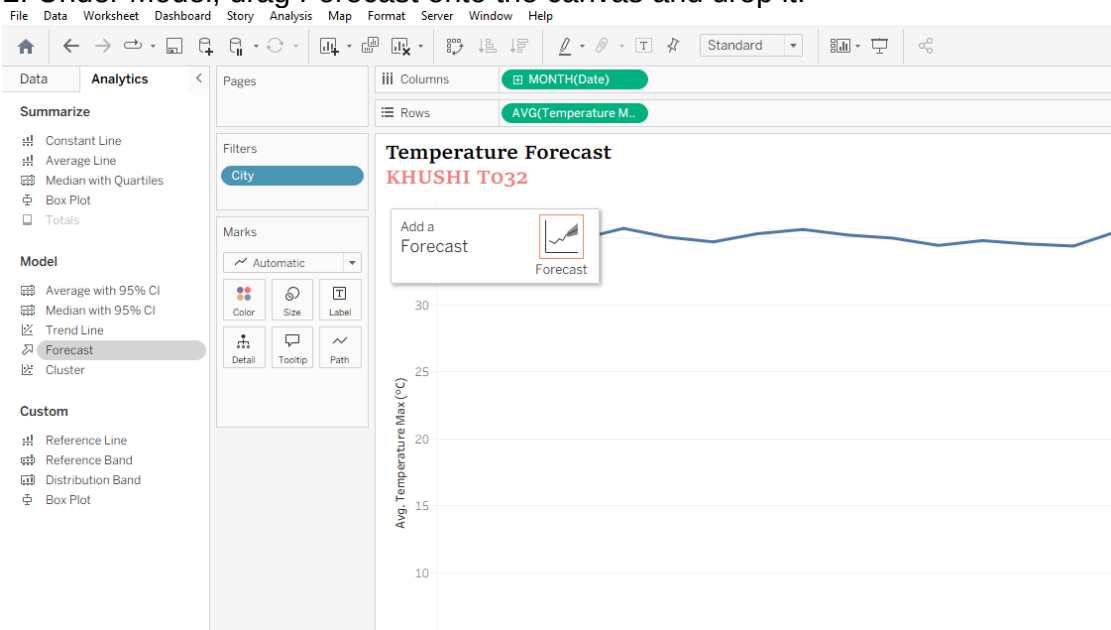
3. Drag Temperature\_Max (°C) to Rows (ensure it is set to AVG(Temperature\_Max)).



4. Drag City to Filters and select 2 or 3 cities (e.g., Mumbai, Delhi, Bengaluru) to keep the view clean.

## 2. Enable Tableau Forecasting

1. Switch to the Analytics Pane on the left sidebar.
2. Under Model, drag Forecast onto the canvas and drop it.



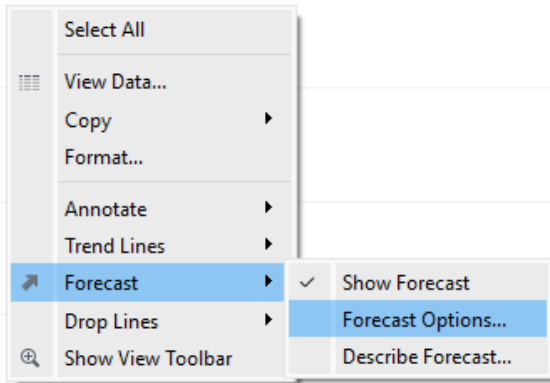
3. Tableau will instantly extend the line chart into the future with a shaded prediction interval band!

## 3. Customize & Analyze the Forecast

1. Right-click anywhere on the chart canvas

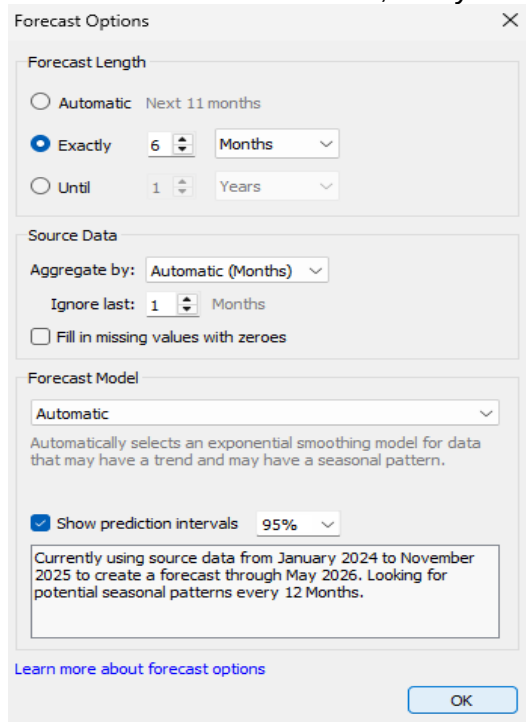
◇ select Forecast

◇Forecast Options....



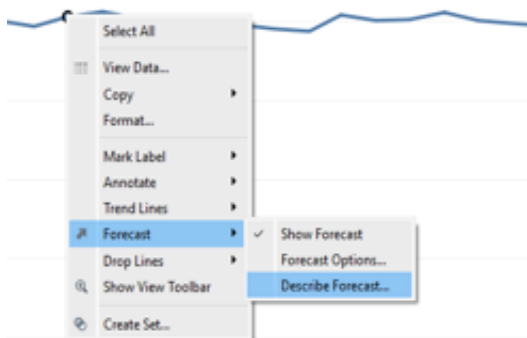
2. Under Forecast Length, set it to Exactly 6 Months into the future.

3. Under Prediction Interval, verify it is set to 95%.



4. To evaluate the mathematical quality of the forecast:

- o Right-click the chart -> Forecast > Describe Forecast....
- o Review the Models tab to check metrics like the Root Mean Squared Error (RMSE) and whether Tableau selected an Additive or Multiplicative model for trend and seasonality.



Describe Forecast ×

Summary Models

**Options Used to Create Forecasts**  
 Time series: Month of Date  
 Measures: Avg. Temperature Max (°C)  
 Forecast forward: 6 months (December 2025 – May 2026)  
 Forecast based on: January 2024 – November 2025  
 Ignore last: 1 month (December 2025)  
 Seasonal pattern: None (Not enough data to search for a seasonal pattern recurring every 12 Months)

**Avg. Temperature Max (°C)**  

| Initial        | Change From Initial      | Seasonal Effect |     | Contribution |        | Quality |
|----------------|--------------------------|-----------------|-----|--------------|--------|---------|
| December 2025  | December 2025 – May 2026 | High            | Low | Trend        | Season |         |
| 35.007 ± 0.882 | 0.000                    | None            |     | 0.0%         | 0.0%   | Ok      |

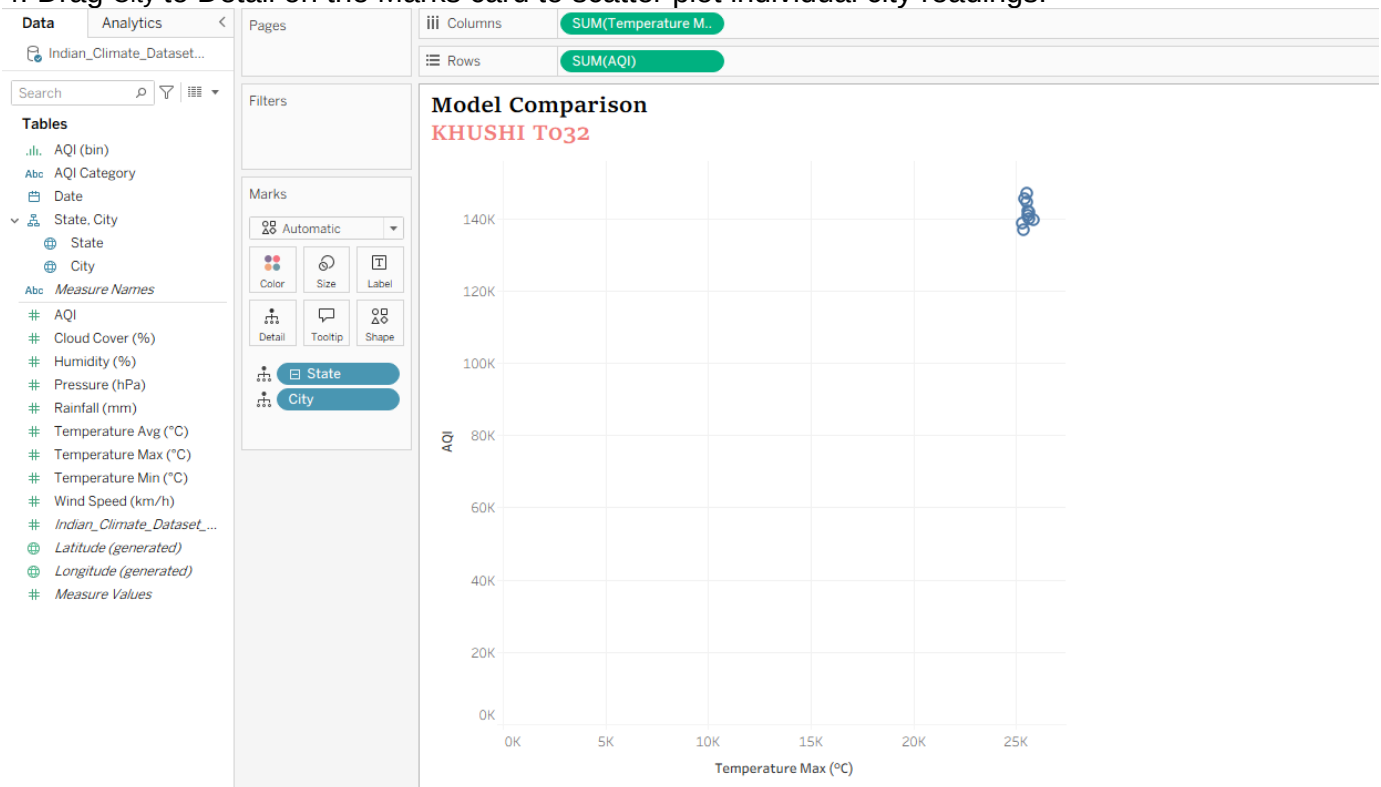
☐ Show values as percentages

Copy to Clipboard [Learn more about the forecast summary](#) Close

## F: Compare different Statistical Models :

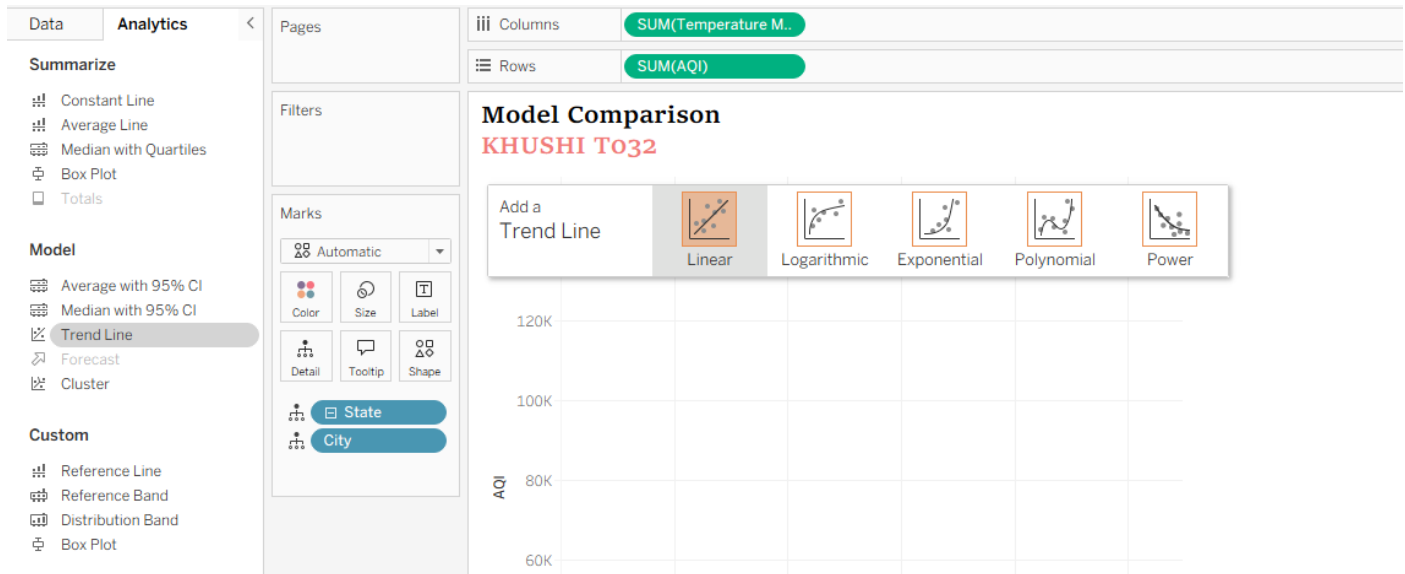
### 1. Set Up the Comparison View

1. Open a new worksheet named Model Comparison.
2. Drag Temperature\_Max (°C) to Columns (set to Continuous Dimension or AVG).
3. Drag AQI to Rows (set to Continuous Dimension or AVG).
4. Drag city to Detail on the Marks card to scatter plot individual city readings.

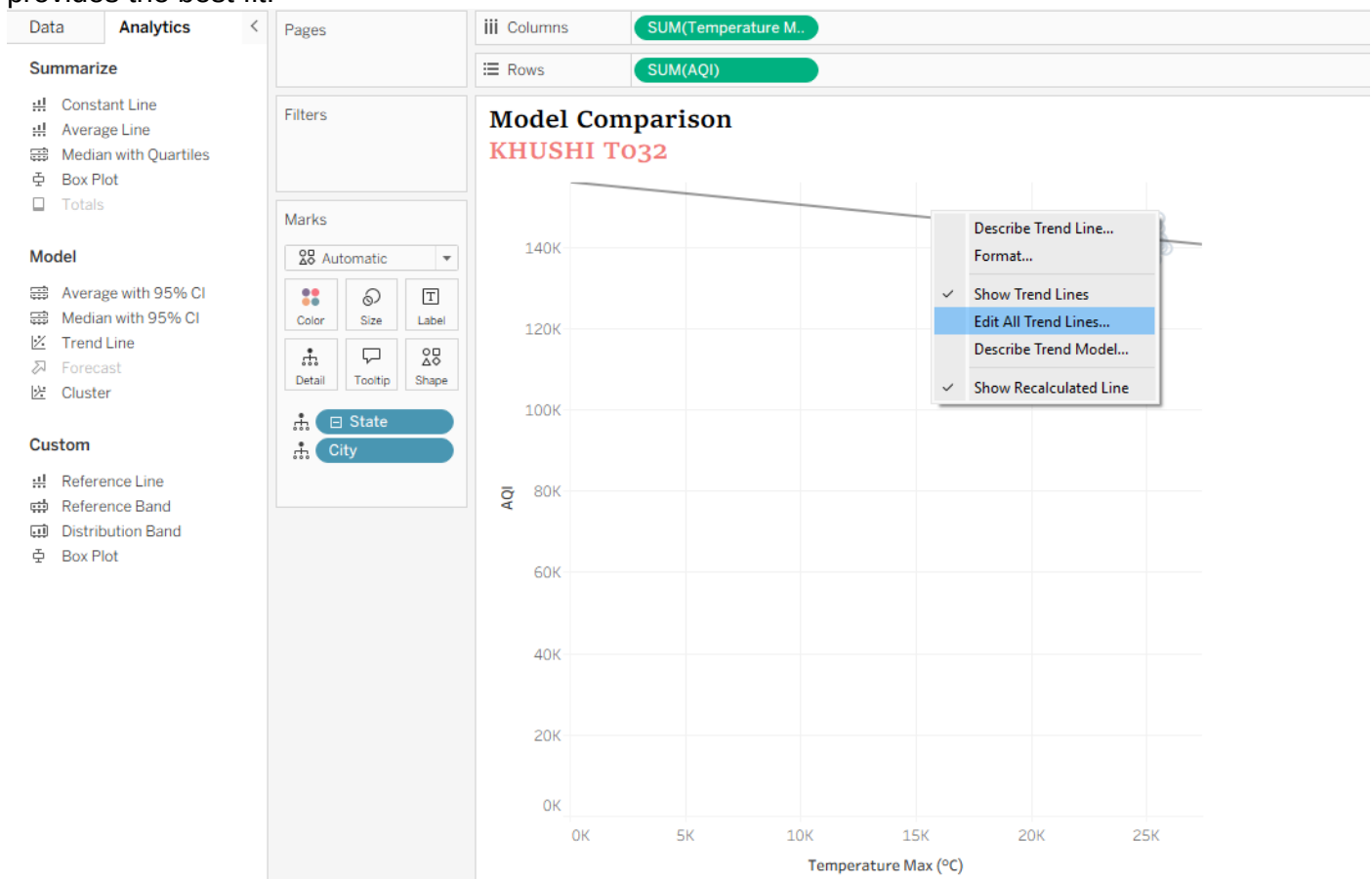


### 2. Add and Swap Statistical Models

1. Open the Analytics Pane on the left.
2. Drag Trend Line onto the canvas and drop it on Linear.



3. Hover over the trend line and note the  $R^2$  and p-value.
4. Right-click the trend line  $\diamond$  Edit Trend Lines....
5. Switch the Model Type through the options:
  - o Linear
  - o Logarithmic
  - o Exponential
  - o Polynomial (Degree 2 or 3): Fits curved relationships.
6. Compare the  $R^2$  value for each model type. The model with the higher  $R^2$  and a p-value  $< 0.05$  provides the best fit.



Trend Lines Options

Model Type

☒ Linear

☐ Logarithmic

☐ Exponential

☐ Power

☐ Polynomial

Degree: 3

Factors

Build separate trend lines based on the following dimensions:

Options

☒ Show tooltips

☐ Show confidence bands

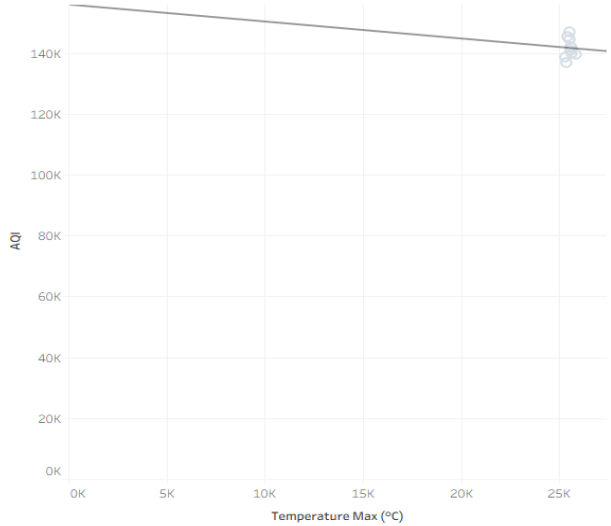
☒ Allow a trend line per color

☒ Show recalculated line for highlighted or selected data points

☐ Force y-intercept to zero

OK

Model Comparison  
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Trend Lines Options

Model Type

☐ Linear

☒ Logarithmic

☐ Exponential

☐ Power

☐ Polynomial

Degree: 3

Factors

Build separate trend lines based on the following dimensions:

Options

☒ Show tooltips

☐ Show confidence bands

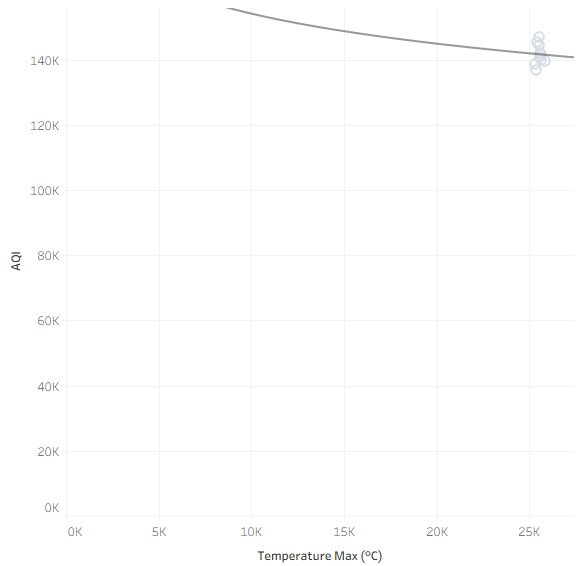
☒ Allow a trend line per color

☒ Show recalculated line for highlighted or selected data points

☐ Force y-intercept to zero

OK

Model Comparison  
KHUSHI T032



Trend Lines Options

Model Type

☐ Linear

☐ Logarithmic

☒ Exponential

☐ Power

☐ Polynomial

Degree: 3

Factors

Build separate trend lines based on the following dimensions:

Options

☒ Show tooltips

☐ Show confidence bands

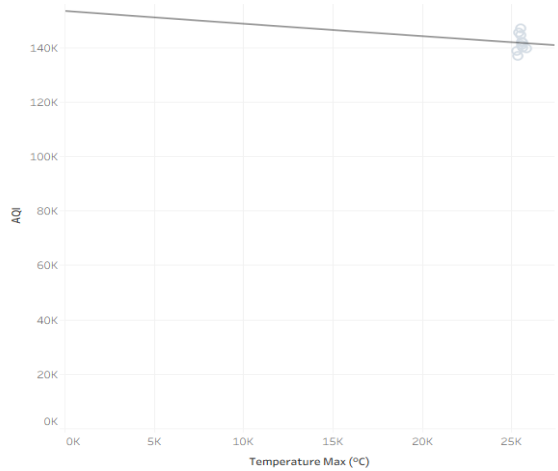
☒ Allow a trend line per color

☒ Show recalculated line for highlighted or selected data points

☐ Force y-intercept to zero

OK

Model Comparison  
KHUSHI T032



Trend Lines Options
✕

**Model Type**

☐ Linear  
☐ Logarithmic  
☐ Exponential  
☐ Power  
☒ Polynomial    Degree: 2

**Factors**

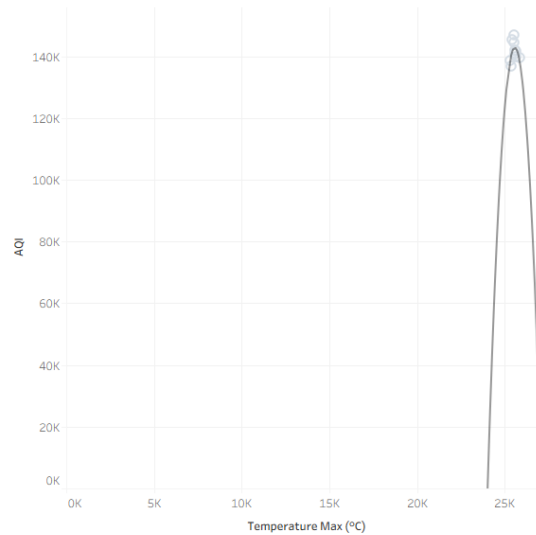
Build separate trend lines based on the following dimensions:

**Options**

☒ Show tooltips  
☐ Show confidence bands  
☒ Allow a trend line per color  
☒ Show recalculated line for highlighted or selected data points  
☐ Force y-intercept to zero

OK

**Model Comparison**  
KHUSHI T032

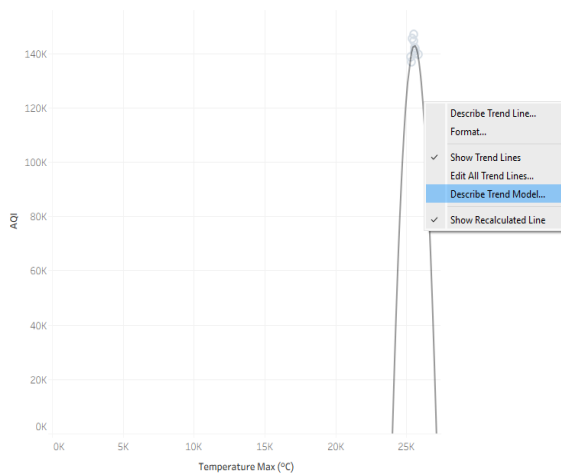


## G: Export and Interpret Statistical Results :

### 1. Export the Statistical Model Description

1. On your Model Comparison worksheet (or any trend/forecast sheet), right-click the trend line or canvas.
2. Select Describe Trend Model... (or Describe Forecast... if on a forecast sheet).

**Model Comparison**  
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**Describe Trend Model**

**Trend Lines Model**

A polynomial trend model of degree 2 is computed for sum of AQI given sum of Temperature Max (°C).

|                                   |                                                               |
|-----------------------------------|---------------------------------------------------------------|
| Model formula:                    | ( Temperature Max (°C)^2 + Temperature Max (°C) + intercept ) |
| Number of modeled observations:   | 10                                                            |
| Number of filtered observations:  | 0                                                             |
| Model degrees of freedom:         | 3                                                             |
| Residual degrees of freedom (DF): | 7                                                             |
| SSE (sum squared error):          | 6.63731e+07                                                   |
| MSE (mean squared error):         | 9.48187e+06                                                   |
| R-Squared:                        | 0.264967                                                      |
| Standard error:                   | 3079.26                                                       |
| p-value (significance):           | 0.340466                                                      |

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